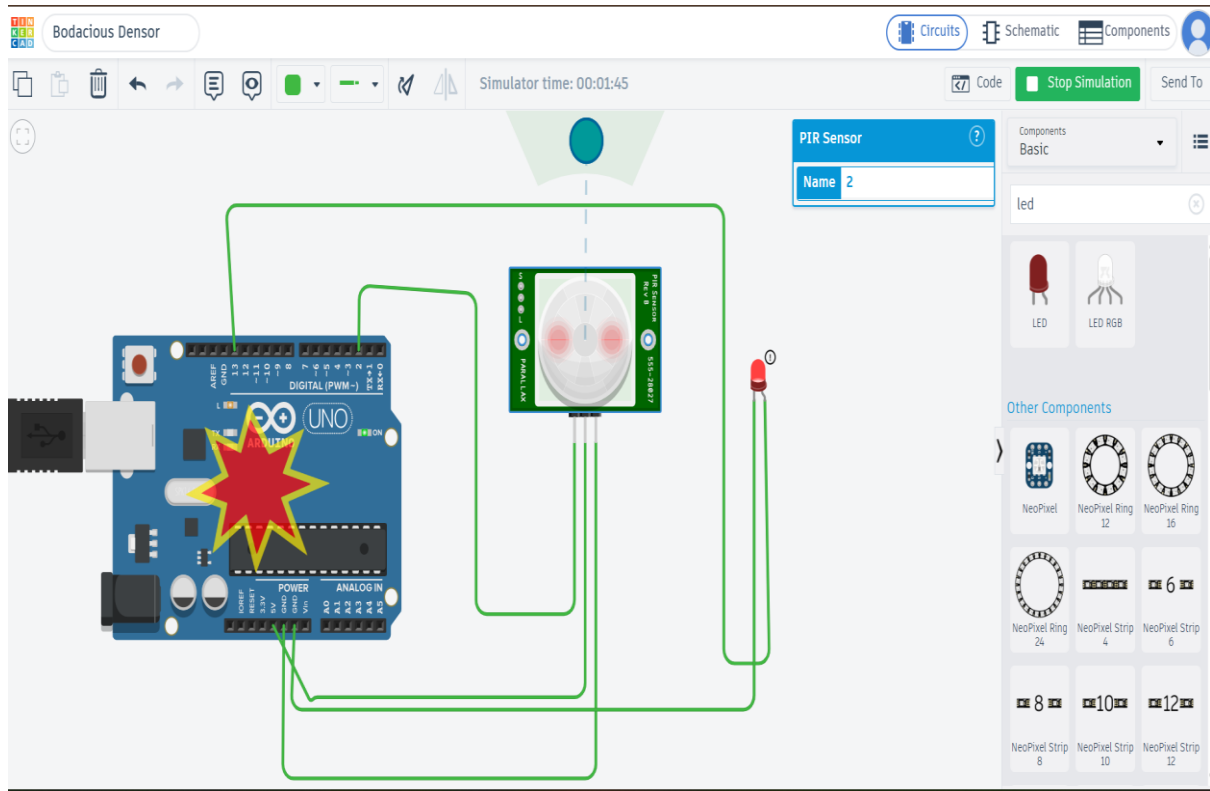


IOT AND EMBEDDED SYSTEMS TASK-2: KANNAN K

PROJECT-MOTION SECURITY SYSTEM USING PIR SENSOR AND LED AND ARDUINO.



SOURCE CODE:

```
const int PIR_PIN = 2; // PIR Sensor signal pin
const int ALARM_PIN = 13; // Buzzer/LED pin

void setup() {
  Serial.begin(9600); // Initialize Serial Monitor
  pinMode(PIR_PIN, INPUT); // Set PIR pin as input
  pinMode(ALARM_PIN, OUTPUT); // Set Alarm pin as output
  Serial.println("System Initialized...");
}

void loop() {
  int motionDetected = digitalRead(PIR_PIN); // Read sensor state
```

```

if (motionDetected == HIGH) {
    digitalWrite(ALARM_PIN, HIGH);    // Turn ON alarm
    Serial.println("ALERT: Intrusion Detected!");
} else {
    digitalWrite(ALARM_PIN, LOW);    // Turn OFF alarm
    Serial.println("STATUS: Safe - No Motion");
}

delay(1000); // 1 second update interval
}

```

Working Explanation: Smart Motion Security System

1. System Overview :

The Smart Motion Security System is an automated embedded project designed to detect human intrusion within a designated zone and immediately trigger defensive output alerts (visual LED and audible buzzer). The system uses an Arduino Uno microcontroller as its processing unit, a PIR (Passive Infrared) Motion Sensor for threat detection, and an LED/Buzzer combination for user feedback.

2. Core Components & Functions:

- **Arduino Uno:** Serves as the brain of the system, running the embedded decision-making logic.
- **PIR Motion Sensor (Pin 2):** Detects changes in infrared radiation emitted by moving warm bodies (e.g., humans or animals) within its field of view.
- **Output Alert (Pin 13):** Activates an LED or Piezo Buzzer to notify nearby users or security personnel of an intrusion.
- **Serial Monitor:** Acts as a real-time logging and monitoring interface, displaying live operational statuses to the terminal.

2. Automation Logic & Decision Making:

The system continuously evaluates input signals from the PIR sensor using an **IF / ELSE** decision structure:

- **State 1: Safe / Normal Operation (ELSE Condition)**
 - When no movement is detected, the PIR sensor sends a **LOW** signal (V_{LOW}) to Digital Pin 2.
 - The Arduino executes the **ELSE** block, maintaining Digital Pin 13 at **LOW** (Alert OFF).
 - **Serial Monitor Output:** Displays "STATUS: Safe - No Motion".
- **State 2: Intrusion / Threat Detected (IF Condition)**

- When an object moves across the sensor's field, the PIR sensor triggers a **HIGH** signal (\$5\text{V}\$) to Digital Pin 2.
- The Arduino evaluates the condition `if (motionDetected == HIGH)` as **TRUE** and executes the **IF** block.
- Digital Pin 13 is set to **HIGH**, turning ON the alarm light/buzzer.
- **Serial Monitor Output:** Displays "ALERT: Intrusion Detected!".

3. Real-World Application:

This mini project simulates real-life smart home security systems, industrial restricted-zone monitoring, and automated safety alarms where immediate local decision-making and live event logging are essential.

